

CURRICULUM VITAE

Hafiz Muhammad Ahsan Shehzad, PhD

Lecturer in Computing | Graph Learning and AI for Health Researcher

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LinkedIn | Google Scholar | WOS: ABQ-5039-2022 | Scopus | ORCID | GitHub | ResearchGate

PROFILE

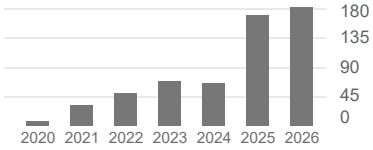
Lecturer in Computing and AI/software engineering researcher with higher-education teaching, curriculum development, and applied research experience across Ireland, China, Pakistan, the United Kingdom, and Australia-linked projects. PhD research at Dalian University of Technology focused on graph transformer models for multi-view brain connectivity analysis and AI-assisted neurological disorder diagnosis. Published in IEEE TNNLS, IEEE JBHI, IEEE TCE, IJCAI, and related venues, with current work spanning graph learning, brain networks, speech-based cognitive impairment detection, and health informatics.

- Teaching: computing, analytics, big data, web programming
- Research: graph transformers for brain and speech intelligence
- Development: Python/PyTorch, full-stack web/mobile, UI/UX

SCHOLAR SNAPSHOT

Cited by

	All	Since 2021
Citations	565	559
h-index	8	8
i10-index	8	8



Google Scholar snapshot checked 19 Sep 2026.

SELECTED ACADEMIC HIGHLIGHTS

Current academic appointment
Lecturer in Computing at Dublin Business School, Ireland, teaching computing, analytics, big data, and web programming.

PhD in Software Engineering
Doctoral research on graph transformers for multi-view brain connectivity analysis and neurological disorder diagnosis.

AI for health research program
Brain graph transformers, speech-based cognitive impairment detection, health informatics, and interpretable graph learning.

Peer-reviewed publication record
IEEE TNNLS, IEEE JBHI, IEEE TCE, IJCAI, IEEE TCSS, IEEE TETCI, Springer/CCIS, and IEEE ICDMW.

RESEARCH INTERESTS

AI for Neuroscience and Neurolinguistics | Brain Science/Health | Health Informatics | Interpretable Graph Learning | Multimodal Learning

EDUCATION

2019 - 24 SEP 2026
Dalian, China

PhD Software Engineering - Dalian University of Technology, China
National key public research university | School of Software | Dalian, China | www.dlut.edu.cn

Thesis: Research on Algorithms for Multi-View Brain Connectivity Analysis Based on Graph Transformers.

Summary: Academic focus in artificial intelligence, software engineering, and computational neuroscience, with emphasis on graph transformers and multi-view brain connectivity analysis for neurological disorder diagnosis.

Award: Chinese Government Scholarship (CSC); Academic Star Award.

2015 - 2018
Islamabad, Pakistan

MS Computer Science - Air University, Pakistan
Federally chartered public-sector university | Department of Computer Science | Islamabad, Pakistan | www.au.edu.pk

Thesis: 3D Joint-based Physiotherapy System for Stroke Rehabilitation with Automatic Assessment and Feedback using Kinect.

Summary: Academic focus in computer vision, human motion analysis, and rehabilitation technology, with emphasis on Kinect-based assessment systems for interactive stroke physiotherapy support.

Award: AU Sponsored Fellowship for MS, Department of Computer Science, Air University, Islamabad.

2011 - 2015
Multan, Pakistan

BS Computer Science - Bahauddin Zakariya University, Pakistan
Public university | Institute of Computing | Multan, Pakistan | www.bzu.edu.pk

FYP: Biometric-based hostel management system using fingerprint recognition.

Summary: Academic focus in programming, software engineering, algorithms, databases, and systems analysis, with emphasis on computer science fundamentals and applied software development.

Award: ICT Scholarship for BS, Ignite National Technology Fund.

TEACHING EXPERIENCE

TEACHING APPOINTMENTS

JAN 2025 - PRESENT
Shenyang, China

Lecturer (Computing) - **Dublin Business School, Ireland**
DBS-SYNU joint programme | College of Artificial Intelligence, Shenyang Normal University | Shenyang, China | Dublin Business School, Ireland | www.dbs.ie

- Deliver DBS-SYNU joint-programme computing modules at the College of Artificial Intelligence, Shenyang Normal University, including analytics, data mining, big data, web programming, and applied software development.
- Develop lectures, labs, assessments, examinations, rubrics, and feedback aligned with DBS academic quality standards and local delivery requirements.
- Use Chinese LMS and teaching platforms, including PTA and Chaoxing, for coursework delivery, assessment management, student engagement, and learning analytics.

AUG 2018 - MAR 2023
Islamabad, Pakistan

Lecturer (Computer Science) - **Air University, Pakistan**
Federally chartered university | Department of Computer Science | Islamabad, Pakistan | www.au.edu.pk

- Taught undergraduate modules across programming, software engineering, databases, web/mobile development, AI, machine learning, computer vision, and data science.
- Developed syllabi, lectures, lab material, assignments, quizzes, examinations, and marking rubrics aligned with course learning outcomes.
- Supervised final-year projects and mentored students in software engineering, AI/ML, web development, and data-oriented project work.

PROFESSIONAL TRAINING

2022 - MAR 2023
Pakistan

Corporate Trainer - **Pakistan Software Export Board (PSEB), Pakistan**
Public-sector IT industry development organization | Industry-Academia Bridge Program | Pakistan | www.pseb.org.pk

- Delivered Advanced Web Deployment (MERN stack) training for the PSEB Industry-Academia Bridge Program.
- Led training-of-trainers (ToT) and student training sessions for top Pakistani universities, including NUST, AU, BU, IIU, FU, and BIT.
- Prepared lecture slides, lab manuals, deployment workflows, coding exercises, project briefs, and practical assessment material for university and industry learners.

JUL 2015 - MAY 2018
Islamabad, Pakistan

Lab Engineer (Computer Science) - **Air University, Pakistan**
Federally chartered university | Department of Computer Science | Islamabad, Pakistan | www.au.edu.pk

- Delivered practical labs for programming, object-oriented programming, databases, data structures, Android/web development, image processing, data science, and Unity-based development.
- Prepared lab manuals, guided debugging and implementation, assessed practical work, and supported course delivery and examination preparation.

RESEARCH EXPERIENCE

2019 - 24 SEP 2026
Dalian, China

Research Associate - **The Alpha Lab, Dalian University of Technology, China**
AI and graph learning research lab | School of Software, Dalian University of Technology | Dalian, China | www.dlut.edu.cn

- Conducted research in graph learning, graph transformers, brain connectivity analysis, health informatics, and AI-assisted neurological disorder diagnosis.
- Developed and evaluated deep learning frameworks for fMRI-based brain network modeling using Python, PyTorch, and experimental validation workflows.
- Collaborated on publications and manuscripts in IEEE TNNLS, IEEE JBHI, IEEE TCE, IJCAI, and related venues.

2015 - 2018
Islamabad, Pakistan

Research Assistant - **Air University, Pakistan**
Federally chartered university | Department of Computer Science | Islamabad, Pakistan | www.au.edu.pk

- Supported applied AI, computer vision, human motion analysis, Kinect-based rehabilitation, gait recognition, and intelligent healthcare-system projects.
- Contributed to prototype development, data processing, experimental evaluation, project documentation, and student research mentoring.

INDUSTRY AND ENTREPRENEURSHIP

2017 - PRESENT
UK / Pakistan

Founder / Technical Lead - **BrandingFlare, Digital Marketing Agency, UK / Pakistan**
Digital technology and marketing services company | Web, mobile, branding, and digital marketing | UK / Pakistan | www.brandingflare.com

- Founded and led a digital technology and marketing services company supporting web, mobile, branding, advertising, and digital marketing projects.
- Managed UI/UX design, full-stack web/mobile development, project scoping, client communication, delivery timelines, and multidisciplinary coordination.

TECHNICAL AND ACADEMIC SKILLS

Education

Curriculum design; lecture and lab development; assessment design; rubrics; moderation; LMS-supported teaching; undergraduate/postgraduate project supervision; professional training delivery.

Research

Problem-oriented research design; proposal development; experimental evaluation; scientific writing; publication preparation; peer-review response; graph transformers; fMRI analysis; multimodal learning.

Development

Python, PyTorch, scikit-learn, pandas, NumPy, data visualization, JavaScript, React, Node.js, Express, MongoDB, REST APIs, UI/UX, full-stack web/mobile application development.

Entrepreneurship

Ideation; feasibility analysis; requirement discovery; service workflow design; project scoping; delivery planning; quality control; financial, operational, and audit-aware workflow management.

CERTIFICATIONS

- Google UX Design Professional Certificate.
- Meta Front-End Developer Professional Certificate.
- Teaching English (TEFL/TESOL), International Open Academy.
- Machine Learning, Stanford University, Coursera.
- Deep Learning Specialization, deeplearning.ai, Coursera.
- Big Data Specialization, University of California San Diego, Coursera.
- Data Science Specialization, University of Michigan, Coursera.
- Game Design and Development Specialization, Michigan State University, Coursera.
- Full-Stack Web Development with React Specialization, Coursera.
- Mastering React, Mosh Hamedani.
- The Ultimate React Native Series, Mosh Hamedani.
- The Complete Node.js Course, Mosh Hamedani.

AWARDS AND FELLOWSHIPS

2025 **Academic Star**, School of International Education, Dalian University of Technology.

2019 **Chinese Government Scholarship (CSC)**, fully funded PhD scholarship.

2015 **AU Sponsored Fellowship for MS**, Air University, Islamabad.

2011 **ICT Scholarship for BS**, Ignite National Technology Fund.

PUBLICATIONS

JOURNAL ARTICLES

J1 **Graph Transformers: A Survey**

A. Shehzad, F. Xia, S. Abid, C. Peng, S. Yu, D. Zhang, K. Verspoor. **IEEE Transactions on Neural Networks and Learning Systems**, 37(7), 3052-3071, 2026. DOI: 10.1109/TNNLS.2025.3646122.

IF 8.9 Q1 CCF-B IEEE Journal

J2 **Dynamic Graph Transformer for Brain Disorder Diagnosis**

A. Shehzad, D. Zhang, S. Yu, S. Abid, F. Xia. **IEEE Journal of Biomedical and Health Informatics**, 29(6), 4388-4400, 2025. DOI: 10.1109/JBHI.2025.3538040.

IF 6.8 Q1 CCF-C EI

J3 **Multiscale Graph Transformer for Brain Disorder Diagnosis**

A. Shehzad, D. Zhang, S. Yu, S. Abid, D. K. Kumar, F. Xia. **IEEE Transactions on Consumer Electronics**, 71(2), 5535-5546, 2025. DOI: 10.1109/TCE.2025.3542794.

IF 10.9 Q1 EI

PUBLICATIONS

JOURNAL ARTICLES, CONTINUED

J4 **An Efficient Pedestrian Detection for Realtime Surveillance Systems Based on Modified YOLOv3**
M. Xu, Z. Wang, X. Liu, L. Ma, A. Shehzad. **IEEE Journal of Radio Frequency Identification**, 6, 972-976, 2022. DOI: 10.1109/JRFID.2022.3212907.
Q1 EI IEEE Journal

J5 **Data-Driven Decision-Making in COVID-19 Response: A Survey**
S. Yu, Q. Qing, C. Zhang, A. Shehzad, G. Oatley, F. Xia. **IEEE Transactions on Computational Social Systems**, 8(4), 1016-1029, 2021. DOI: 10.1109/TCSS.2021.3075955.
Q1 EI IEEE Transactions

J6 **Matching Algorithms: Fundamentals, Applications and Challenges**
J. Ren, F. Xia, X. Chen, J. Liu, M. Hou, A. Shehzad, N. Sultanova, X. Kong. **IEEE Transactions on Emerging Topics in Computational Intelligence**, 5(3), 332-350, 2021. DOI: 10.1109/TETCI.2021.3067655.
Q1 EI IEEE Transactions

JOURNAL ARTICLES UNDER REVIEW

UR1 **mBrainGT: Modular Graph Transformer for Brain Disorder Diagnosis**
A. Shehzad, S. Yu, D. Zhang, S. Abid, X. Cheng, J. Zhou, F. Xia. **IEEE Transactions on Neural Networks and Learning Systems**, under review, 2026. Preprint DOI: 10.1101/2024.08.30.24312819.
IF 8.9 Q1 CCF-B

UR2 **SpeechGT: Multimodal Graph Transformer for Early Detection of Alzheimer's Disease Using Spontaneous Speech**
S. Abid, D. Zhang, A. Shehzad, S. Yu, F. Xia. **IEEE/ACM Transactions on Audio, Speech, and Language Processing**, under review, 2026.
Q1 CCF-B

UR3 **Modeling Dynamic Multimodal Interactions for Cognitive Impairment Detection from Speech**
S. Abid, D. Zhang, A. Shehzad, F. Xia. **IEEE Journal of Biomedical and Health Informatics**, under review, 2026.
IF 6.8 Q1 CCF-C

UR4 **SpeechCGT: A Knowledge-Guided Causal Graph Transformer for Interpretable Cognitive Impairment Detection from Multimodal Speech**
S. Abid, D. Zhang, A. Shehzad, F. Xia. **Transactions of the Association for Computational Linguistics**, under review, 2026.
Q1 CCF-B

CONFERENCES

C1 **BrainCGT: A Brain Graph Transformer for Modeling Causal Connectivity in Neurological Disorder Diagnosis**
A. Shehzad, D. Zhang, S. Abid, S. Yu, X. Zheng, H. Lin, F. Xia. **Proceedings of the 35th International Joint Conference on Artificial Intelligence (IJCAI 2026)**, 6509-6517, 2026. DOI: 10.24963/ijcai.2026/724.
CCF-B CORE A* IJCAI

C2 **SpeechHGT: A Multimodal Hypergraph Transformer for Speech-Based Early Alzheimer's Disease Detection**
S. Abid, D. Zhang, A. Shehzad, J. Ren, S. Yu, H. Lin, F. Xia. **34th International Joint Conference on Artificial Intelligence (IJCAI 2025)**, 9131-9139, 2025. DOI: 10.24963/ijcai.2025/1015.
CCF-A IJCAI

C3 **Knowledge Graph Completion via Subgraph Topology Augmentation**
H. Huang, F. Ding, F. Zhang, Y. Wang, C. Peng, A. Shehzad, Q. Lei, L. Cong, S. Yu. **Social Media Processing (SMP 2023), Springer CCIS**, 14-29, 2023. DOI: 10.1007/978-981-99-7596-9_2.
EI Springer CCIS

C4 **Cross Network Representation Matching with Outliers**
M. Hou, J. Ren, F. G. Febrinanto, A. Shehzad, F. Xia. **IEEE International Conference on Data Mining Workshops (ICDMW 2021)**, 951-958, 2021. DOI: 10.1109/ICDMW53433.2021.00124.
EI IEEE Conference

SELECTED R&D PROJECTS

SELECTED R&D PROJECTS, CONTINUED

01 **Scale-Aware Brain Connectivity Analysis Based on Multiscale Graph Transformer - The Alpha Lab, China**

Role: Key Researcher **Funding:** National Natural Science Foundation of China, Grant No. 62576073

Description: Developed BrainMGT, a multiscale brain graph transformer for neurological disorder diagnosis using hierarchical spatial and temporal brain connectivity. The framework models brain networks at micro-, meso-, and macro-scales and integrates intra-scale and inter-scale dependencies through self-attention and cross-attention.

Data and Tools: fMRI-based brain connectivity; ADNI, PPMI, and ABIDE datasets; graph transformers; multiscale brain networks; Python/PyTorch; attention-based modeling.

02 **State-Aware Brain Connectivity Analysis Based on Dynamic Graph Transformer - The Alpha Lab, China**

Role: Key Researcher **Funding:** National Natural Science Foundation of China, Grant No. 62576075

Description: Developed BrainDGT, a dynamic graph transformer for brain disorder diagnosis that estimates adaptive brain states through HRF deconvolution and models localized functional brain modules. The framework addresses limitations of fixed sliding-window dynamic networks and improves temporal sensitivity in fMRI-based diagnosis.

Data and Tools: fMRI data; ADNI, PPMI, and ABIDE datasets; HRF deconvolution; dynamic graph construction; graph attention; self-attention; adaptive fusion; Python/PyTorch.

03 **Direction-Aware Brain Connectivity Analysis Based on Causal Graph Transformer - The Alpha Lab, China**

Role: Key Researcher **Funding:** National Natural Science Foundation of China, Grant No. 62076051

Description: Developed BrainCGT, a causal brain graph transformer that models directed interactions between brain regions for neurological disorder diagnosis. The model uses causal connectivity and direction-aware attention to capture asymmetric information flow and improve interpretability of disease-related brain mechanisms.

Data and Tools: fMRI time series; ADNI, PPMI, and ABIDE datasets; multivariate Granger causality; directed graph modeling; modular brain networks; graph transformer; direction-aware attention.

04 **SpeechHGT: Multimodal Hypergraph Transformer for Speech-based Alzheimer's Disease Detection - The IR Lab, China**

Role: Key Researcher **Funding:** National Language Commission Key Program, Grant No. ZDI145-80

Description: Developed a multimodal hypergraph transformer for early Alzheimer's disease detection from spontaneous speech. The model represents linguistic and acoustic features as hypergraphs and captures pairwise and higher-order feature interactions for improved non-invasive diagnostic screening.

Data and Tools: DementiaBank speech datasets; acoustic and linguistic features; multimodal hypergraphs; hypergraph attention; transformer-based classification; Python/PyTorch.

05 **SpeechDGT: Dynamic Speech Graph Transformer for Cognitive Impairment Detection - The IR Lab, China**

Role: Key Researcher **Funding:** Fundamental Research Funds for the Central Universities, Grant No. DUT24RW213

Description: Developed a dynamic speech graph transformer that models spontaneous speech as temporally evolving acoustic-linguistic interactions. The framework encodes short-window speech interactions, abstracts them into latent cognitive moments, and uses moment-aware attention for interpretable process-level impairment detection.

Data and Tools: DementiaBank Cookie Theft corpora; Pitt, Delaware, and WLS cohorts; acoustic and linguistic features; feature-level graph attention; temporal abstraction; moment-aware transformer; Python/PyTorch.

06 **SpeechCGT: Knowledge-Guided Causal Graph Transformer for Cognitive Impairment Detection - The IR Lab, China**

Role: Key Researcher **Funding:** Liaoning Provincial Federation of Social Sciences Economic and Social Development Research Project, Grant No. 2025lslybkt-064

Description: Developed a knowledge-guided causal graph transformer that organizes multimodal speech biomarkers into a neurolinguistic hierarchy of speech features, cognitive functions, and cognitive domains. The framework supports both cognitive impairment classification and interpretable cognitive-pathway mapping.

Data and Tools: DementiaBank Pitt and Delaware corpora; held-out Wisconsin Longitudinal Study corpus; acoustic and transcript-derived biomarkers; causal graph transformer; neurolinguistic prior graph; pathway-gated attention.

07 **Energy Data Analysis and Visualization Project - C4NET, Federation University Australia, Australia**

Role: Project Supervisor **Funding:** Centre for New Energy Technologies, Federation University Australia, Institutional Funding

Description: Supported data-driven analysis for modern energy-system planning by identifying useful energy-sector datasets, producing visualizations and reports, and improving access to interpretable data for technical and non-technical stakeholders.

Data and Tools: Energy-sector datasets; data cleaning; exploratory analysis; dashboards/visualizations; reporting workflows.

08 **HL7-based Medication Chart - University of South Australia, Australia**

Role: Project Supervisor **Funding:** University of South Australia, Institutional Funding

Description: Designed and developed a medication-chart tool to support medication prescribing, scheduling, and tracking, particularly for elderly-care contexts. Contributed across UI/UX design and full-stack development while incorporating HL7 FHIR concepts and health information integration requirements.

Data and Tools: HL7 FHIR concepts; healthcare workflow modeling; UI/UX design; full-stack development; medication scheduling and tracking logic.

09 **Anomaly Detection in Electronic Health Records - University of South Australia, Australia**

Role: Project Supervisor **Funding:** University of South Australia, Institutional Funding

Description: Developed a concept system for identifying anomalous patterns in healthcare records, including diagnosis, treatment, and prescription behavior. The project connected machine learning workflows with real-world EHR data challenges in health informatics.

Data and Tools: MIMIC-III and MIMIC-IV electronic health record datasets; anomaly detection; healthcare data preprocessing; machine learning model development.

10 **Smart Physiotherapy System for Hemiplegic Stroke Rehabilitation - Air University and Shifa International Hospital, Pakistan**

Role: MS Researcher **Funding:** Air University and Shifa International Hospital, Joint Institutional Funding

Description: Developed a markerless motion-capture rehabilitation system using Microsoft Kinect and augmented-reality concepts to guide, assess, and provide feedback on range-of-motion exercises for hemiplegic stroke rehabilitation. The system supported active and active-assisted ROM exercises and was evaluated with physiotherapy input.

Data and Tools: Microsoft Kinect; 3D joint tracking; range-of-motion exercise assessment; augmented feedback; rehabilitation workflow design.

11 **Gait Recognition using Kinect - Air University, Pakistan**

Role: Project Supervisor **Funding:** Air University, Institutional Funding

Description: Implemented a Kinect-based gait recognition approach using skeleton trajectories and joint movement features. The project extracted horizontal and vertical distance features across gait cycles to support person-identification and movement-analysis tasks.

Data and Tools: Microsoft Kinect; skeleton tracking; gait cycle analysis; horizontal and vertical distance features; machine learning classification.

12 **KinectiRehab: Serious Game for Remote Physiotherapy - Air University, Pakistan**

Role: Project Supervisor **Funding:** Air University, Institutional Funding

Description: Developed a serious-game rehabilitation framework using Microsoft Kinect to capture 3D exercise data and transmit it to a web application for physiotherapist review, remote monitoring, and assessment support.

Data and Tools: Microsoft Kinect; serious-game interaction; web application; exercise data capture; remote rehabilitation monitoring.

13 **Sign Language Recognition - Air University, Pakistan**

Role: Project Supervisor **Funding:** Air University, Institutional Funding

Description: Developed a computer-vision project for interpreting sign-language gestures through visual feature extraction and classification. The work contributed to applied human-computer interaction and accessibility-oriented recognition systems.

Data and Tools: Image/video gesture data; computer vision; feature extraction or deep learning pipeline; classification model.

ACADEMIC SERVICE AND MEMBERSHIPS

ACADEMIC SERVICE

- PC Chair, WWW 2026 Semantics & Knowledge Track.
- Reviewer, ACM Transactions on Intelligent Systems and Technology.
- Reviewer, ACM Transactions on Knowledge Discovery from Data.
- Reviewer, IEEE Access.
- Reviewer, IEEE Journal of Biomedical and Health Informatics.
- Reviewer, IEEE Transactions on Medical Imaging.
- Reviewer, IEEE Transactions on Neural Networks and Learning Systems.

PROFESSIONAL MEMBERSHIPS

- Member, Artificial Intelligence Association of Ireland (AIAI), 2026-present.
- Member, European Association for Artificial Intelligence (EurAI), through AIAI affiliation, 2026-present.

LANGUAGES

English
Fluent

Urdu
Fluent / native

Punjabi
Fluent

Chinese
Basic

ACADEMIC AND PROFESSIONAL REFERENCES

Dr. David Williams
Academic Director in Computing, Dublin Business School, Ireland.
Relationship: Current Academic Director
david.williams@dbs.ie

Prof. Feng Xia
Professor, School of Computing Technologies, RMIT University, Australia.
Relationship: PhD Supervisor
f.xia@rmit.edu.au

Prof. Dongyu Zhang
Professor, School of Software Technology, Dalian University of Technology, China.
Relationship: PhD Supervisor
zhangdongyu@dlut.edu.cn

Prof. Shuo Yu
Associate Professor, School of Computer Science and Technology, Dalian University of Technology, China.
Relationship: PhD Co-Supervisor
yushuo@dlut.edu.cn

Prof. Atta ur Rehman Khan
Professor, College of Engineering and IT, Ajman University, United Arab Emirates.
Relationship: Academic Mentor
a.khan@ajman.ac.ae

Prof. Ehsan Ahmad
Associate Professor, College of Computing and Informatics, Saudi Electronic University, Saudi Arabia.
Relationship: Academic Mentor
e.ahmad@seu.edu.sa